

Creator / Influencer Module – Backend Technical Design (High Level)

1. Stack and conventions (unchanged)

- **Stack:** Flask + SQLAlchemy, existing `User`/role system, JWT, Blueprints.
 - **Pattern:** Routes → Controllers/Services → Models; reuse `common.database`, `common.response`, `common.decorators`, and existing auth/decorators.
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2. Auth and role

- **Extend `UserRole`** in `auth/models/models.py`: add `CREATOR = 'creator'`.
 - **Creator signup:** New signup path (e.g. `/api/auth/signup/creator`) that creates `User(role=UserRole.CREATOR)` and triggers creator onboarding (no `MerchantProfile`).
 - **Creator login:** Use same auth flow; after login, if `role == CREATOR` redirect to creator dashboard (handled by frontend; backend only returns role).
 - **Guards:** Add `creator_role_required` (and/or extend `role_required`) so creator-only routes allow only `UserRole.CREATOR`; reuse existing JWT and `get_jwt_identity()`.
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3. New and extended data (high level)

Area	What	Purpose
Creator profile	New model <code>CreatorProfile</code> (1:1 with <code>User</code>)	Categories (M2M with <code>Category</code> , min 5), portfolio (links/samples), availability flag, optional language prefs.
Campaign / Deal	New model <code>Campaign</code>	Links merchant, product, creator; status lifecycle (draft → sent → accepted → active → submitted → approved → live → completed); terms (commission %, optional cap qty, window); human-friendly <code>campaign_code</code> ; deliverables (e.g. “1 reel” in V1).
Reel	Extend existing <code>Reel</code> (or add optional link table)	Associate creator-uploaded reels to a campaign: e.g. <code>campaign_id</code> (FK), <code>creator_id</code> (FK to <code>users.id</code>). For creator reels, “merchant” is implied by campaign; approval flow (pending → approved/rejected by merchant).
Attribution		

	New model <code>ReelProductClick</code> (or <code>AttributionEvent</code>)	Store “user clicked product from this reel” (user_id, reel_id, product_id, campaign_id, clicked_at) for last-touch attribution within X days.
Settlement	New models: e.g. <code>AttributedSale</code> , <code>CreatorSettlement</code> <code>Ledger</code>	Which order items count as attributed to which campaign; ledger rows for creator earning and AOIN fee (pending → eligible → paid / reversed).

4. Core flows (backend responsibilities)

- **Creator onboarding (post signup):**
 - API to create/update `CreatorProfile`: set categories (validate ≥ 5), portfolio, availability.
 - Categories stored as M2M (e.g. `creator_category` table: creator_profile_id, category_id).
- **Merchant “find creators”:**
 - List creators filtered by category (match product’s category), availability, optional portfolio.
 - Returns creator list (id, name, categories, portfolio preview, availability) — no product/stock.
- **Merchant “create campaign”:**
 - Input: product_id, creator_id, terms (commission %, optional cap qty, window), deliverables.
 - Create `Campaign` in draft → “send” sets status to sent; creator sees it in “offers”.
- **Creator “accept / reject deal”:**
 - Accept → campaign status to accepted/active; Reject → rejected.
 - Only assigned creator can accept/reject.
- **Creator “upload reel”:**
 - Allowed only for campaigns in “active” (and in the right status for “submitted”).
 - Upload video (reuse existing reel upload pipeline, e.g. S3/Cloudinary); create/update `Reel` with `campaign_id`, `creator_id`, product from campaign; status = pending merchant approval.
- **Merchant “approve reel”:**
 - Approve → reel goes live (visible in feed); Reject → with reason, creator can re-upload.
 - Reel visibility logic should consider this approval state for creator reels.
- **Attribution (reel → product click):**
 - When user opens product from a reel (app sends reel_id + product_id), backend records `ReelProductClick` (user_id, reel_id, product_id, campaign_id, timestamp).
 - Optional: cap one “last click” per (user, product) in a sliding window (e.g. keep only latest within 7 days) to simplify last-touch.
- **Order → attribution:**

- On order placement (or when order becomes “delivered”), for each order line (product_id), check if buyer has a recent `ReelProductClick` for that product within attribution window (e.g. 7 days); if yes, attribute that order item to the corresponding campaign (last-touch).
 - Create/update `AttributedSale` (order_id, order_item_id, campaign_id, creator_id, sale amount or net amount).
 - Fraud rule: do not attribute if `order.user_id` is the creator or the merchant of that campaign.
 - **Settlement and payouts:**
 - When order is delivered and return window has passed, mark attributed sales as “eligible”.
 - Periodically (or on event) create ledger entries: creator earning (commission on net), AOIN fee (e.g. 5% of attributed net).
 - Support reversal on refund/return (ledger reversal or negative entry).
 - Payout run (weekly/monthly or threshold-based) reads “eligible” creator balances and triggers payouts (integration with payout provider later); mark as “paid” and record in ledger.
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5. APIs (grouped by domain)

- **Auth:**
 - `POST /api/auth/signup/creator`
 - (Existing login; role in token/response for redirect.)
- **Creator:**
 - Onboarding: `GET/PUT /api/creator/profile` (categories, portfolio, availability).
 - Dashboard: `GET /api/creator/deals` (offers + active + completed), `GET /api/creator/earnings` (pending, eligible, paid).
 - Reel: `POST /api/creator/reels` (select campaign, upload video, description); `GET /api/creator/reels` (my reels per campaign).
 - Deals: `POST /api/creator/deals/<id>/accept`, `POST /api/creator/deals/<id>/reject`.
- **Merchant:**
 - Under existing merchant prefix (e.g. `/api/merchant-dashboard`):
 - `GET /api/merchant-dashboard/creators` (query: category_ids, availability).
 - `POST /api/merchant-dashboard/campaigns` (create), `GET /api/merchant-dashboard/campaigns` (list/filter), `GET/PATCH /api/merchant-dashboard/campaigns/<id>`.
 - `POST /api/merchant-dashboard/campaigns/<id>/reels/<reel_id>/approve`, `POST .../reject` (with reason).
 - `GET /api/merchant-dashboard/campaigns/<id>/attributed-sales` (or summary).
- **Public / app (attribution):**

- `POST /api/reels/<reel_id>/product-click` (body: `product_id`; user from JWT or session) → creates attribution event.
 - **Reel feed:**
 - Existing reel feed can be extended to include creator reels (with `campaign_id/creator_id`) and product link; product page or reel card calls the product-click API above.
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6. Integration with existing code

- **User:** Add `CreatorProfile` relationship (like `merchant_profile`, `customer_specific_profile`).
 - **Reel:** Add nullable `campaign_id`, `creator_id`; for creator reels, `merchant_id` can come from campaign or be duplicated for current visibility logic; keep existing `approval_status` and use it for “merchant approval” for creator reels.
 - **Order / OrderItem:** No schema change; attribution is in `AttributedSale` / ledger keyed by `order_id` and `order_item_id`.
 - **Categories:** Reuse `Category` for creator categories and for “find creators by product category”.
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7. Suggested implementation order (backend)

1. `UserRole.CREATOR` + creator signup/login + `CreatorProfile` (with categories M2M).
 2. Campaign model + status lifecycle + merchant APIs (create campaign, list, send offer).
 3. Creator deal APIs (list offers, accept/reject).
 4. Reel extension (`campaign_id`, `creator_id`) + creator upload reel API + merchant approve/reject reel.
 5. Attribution: `ReelProductClick` + product-click API + order attribution job (last-touch, 7-day window, fraud rule).
 6. Settlement: `AttributedSale` + `CreatorSettlementLedger` + eligibility (delivered + return window) + reversal on refund.
 7. Creator earnings APIs (pending/eligible/paid) and merchant campaign attributed-sales summary.
 8. (Later) Payout integration and admin moderation hooks.
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8. Ecommerce (frontend) – high level

- **Creator app (or web):** Creator signup → onboarding (5 categories, portfolio, availability) → dashboard (deals, upload reel, earnings).
- **Merchant dashboard:** New “Creator Campaigns” section: find creators, create campaign, manage campaigns, approve/reject reels, view attributed sales/commission summary.
- **Customer app:** Reel feed includes creator reels; “open product” from reel triggers product-click API; checkout/order unchanged.